

HENDRIK N. SORENSEN

Ann Arbor, Michigan 48104, United States

Hnsorens@umich.edu | 616-840-5405 | hnsorens.github.io | linkedin.com/in/hendrik-sorensen/

EDUCATION

University of Michigan, College of Engineering

Start: August 2024 | Expected Graduation: May 2028

Bachelor of Science in Computer Engineering; Minor in Physics

Cumulative GPA: 3.6

Coursework: Data Structures & Algorithms, Discrete Math, Linear Algebra, Calculus-Based Physics, Circuit Analysis

SKILLS

React, JavaScript, TypeScript, HTML, CSS, Material-UI, Electron, REST API, CQRS, ZMQ, Unity, Unreal Engine, C#, MRTK, HoloLens2, Vulkan, OpenGL, GLSL, GLFW, C, C++, GDB, GCC, G++, UEFI/EFI, OS Development, Java, Git, Playwright, Unit Testing, Debugging, Problem Solving, Data Structures, Agile

EXPERIENCE

Jovian Software Consulting LLC | Grand Rapids, Michigan

Software Engineer Intern | May 2025 – Aug 2025 | (In-Person)

- Enhanced GoJS aerospace modeling system in a commercial electron app using TypeScript and HTML, transforming it from limited functionality to feature-rich while reducing code duplication by 60 percent
- Reduced errors in refactor by 90 percent through comprehensive Playwright test suites
- Applied CQRS pattern to Java REST APIs and validated compatibility with Swagger to eliminate all incompatibilities

Software Engineer Intern | Jun 2022 – Aug 2024 | (In-Person; Summer and Part-time)

- Developed AR mission planning app using (C#), Unity, MRTK, and XR to create interactive terrain visuals, gesture-based UX with HoloLens 2
- Migrated AR app from Unity to Unreal Engine (C++) and constructed a terrain with Cesium, improving HoloLens 2 performance and enabling real-time vehicle position and detonation updates from ZMQ messages
- Implemented terrain system using Blender and Cesium with AR locking functionality for stable holographic placement
- Adopted Agile, Git, and Unity Profiler to refine hologram performance, achieving 90 percent spatial accuracy

FIRST Robotics Competition | COMETS 3357 | Forest Hills, Michigan

Team Captain, Software Lead, Drive Team | May 2021 – Aug 2024

- Directed team strategy and assigned tasks for a 30-person team, mentoring underclassmen in engineering and software principles to ensure project success
- Fabricated the C++ codebase, computer vision system (OpenCV/AprilTags) for localization and autonomous navigation
- Won five district competitions and performed well at world championships in Houston, Texas, in 2022, 2023, and 2024
- Mentored the middle school robotics team in programming, robot design, and machining, while serving as their drive coach

PROJECTS

Unix Operating System

- Designed and built a 64-bit POSIX-compliant OS from scratch, implementing core functionalities including virtual memory with paging, multitasking, an ELF loader, an ext2 filesystem, and a system call interface to support user-space processes
- Developed the entire low-level software stack for a custom operating system, including a UEFI bootloader, hardware abstraction layer, interrupts, virtual filesystem (VFS), custom shell, process scheduler, and a from-scratch C standard library

Assembler

- Formulated an x86-64 assembler in C with ELF64 binary generation, implementing lexical analysis, symbol resolution, and direct machine code encoding for more than 38 instructions
- Designed a four-stage assembler pipeline (lexical analyzer/parser/linker/code generation) producing bit-accurate Linux executables, validated against more than 842,000 test cases

Vulkan Voxel Engine

- Pioneered a real-time ray-traced voxel engine using Vulkan's hardware acceleration, achieving 60 FPS with over 1 billion bit-packed voxels through multi-threaded chunk generation and GPU-optimized 3D texture storage
- Engineered a custom ray-marching pipeline with LOD-aware traversal in GLSL, implementing atomic lighting accumulation and multi-pass rendering for global illumination effects